

Enterprise Database Infrastructure Administration Lab

Design, Deployment, Security Hardening, Backup, Replication and
Monitoring of PostgreSQL Infrastructure on Linux

David Gottlieb SITTI
<https://www.linkedin.com/in/davidgottliebsitti/>

SUMMARY

POSTGRESQL DEPLOYMENT AND INITIAL ADMINISTRATION	2
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POSTGRESQL DEPLOYMENT AND INITIAL ADMINISTRATION

I. Objective

The purpose of this deliverable is to deploy and configure the primary PostgreSQL database server that will serve as the foundation of the enterprise database infrastructure.

- At the end of this phase, you will have :
- PostgreSQL installed and operational
- Remote administration enabled
- Enterprise database structure created
- Administrative roles configured
- Authentication configured
- Initial data validation completed
- Documentation-ready evidence collected

II. PostgreSQL Overview

What is PostgreSQL ?

PostgreSQL is an enterprise-grade open-source Relational Database Management System (RDBMS) widely used for :

- Banking systems
- Enterprise applications
- ERP solutions
- Cloud-native applications
- Data analytics platforms

The PostgreSQL server process is responsible for :

- Data storage
- Transaction processing
- Authentication
- Query execution
- Backup operations
- Replication

III. Verify System Readiness

On pg-primary :

- Verify resources :
 - free -h
 - df -h
 - hostnamectl
- Verify connectivity
 - ping -c 4 db-admin
 - ping -c 4 monitoring

```

root@pg-primary:~# df -h
Filesystem                Size      Used Avail Use% Mounted on
tmpfs                     387M        1,3M  386M   1% /run
/dev/mapper/ubuntu--vg-ubuntu--lv 19G        6,6G   12G  38% /
tmpfs                     1,9G         0   1,9G   0% /dev/shm
tmpfs                     5,0M         0   5,0M   0% /run/lock
/dev/sda2                 2,0G       104M   1,7G   6% /boot
tmpfs                     387M       12K   387M   1% /run/user/1000
root@pg-primary:~# free -h
               total        used        free      shared  buff/cache   available
Mem:            3,8Gi         485Mi        2,6Gi         1,3Mi         918Mi        3,3Gi
Swap:           3,8Gi           0B         3,8Gi
root@pg-primary:~# hostnamectl
Static hostname: pg-primary
Icon name: computer-vm
Chassis: vm
Machine ID: f65b5af74fef4148a482b651398724e0
Boot ID: 03a9a0ee5f2a48a9aae2590e1f29b1f9
Virtualization: vmware
Operating System: Ubuntu 24.04.4 LTS
Kernel: Linux 6.8.0-124-generic
Architecture: x86_64
Hardware Vendor: VMware, Inc.
Hardware Model: VMware Virtual Platform
Firmware Version: 6.00
Firmware Date: Mon 2025-03-24
Firmware Age: 1y 2month 2w 5d
root@pg-primary:~# ping -c 4 db-admin
PING db-admin (192.168.10.10): 56(84) bytes of data:
64 bytes from db-admin (192.168.10.10): icmp_seq=1 ttl=64 time=0.553 ms
64 bytes from db-admin (192.168.10.10): icmp_seq=2 ttl=64 time=0.939 ms
64 bytes from db-admin (192.168.10.10): icmp_seq=3 ttl=64 time=0.785 ms
64 bytes from db-admin (192.168.10.10): icmp_seq=4 ttl=64 time=0.900 ms

--- db-admin ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3011ms
rtt min/avg/max/mdev = 0.553/0.794/0.939/0.150 ms
root@pg-primary:~# ping -c 4 monitoring
PING monitoring (192.168.10.30): 56(84) bytes of data:
64 bytes from monitoring (192.168.10.30): icmp_seq=1 ttl=64 time=0.822 ms
64 bytes from monitoring (192.168.10.30): icmp_seq=2 ttl=64 time=0.875 ms
64 bytes from monitoring (192.168.10.30): icmp_seq=3 ttl=64 time=0.929 ms
64 bytes from monitoring (192.168.10.30): icmp_seq=4 ttl=64 time=0.872 ms

--- monitoring ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3005ms
rtt min/avg/max/mdev = 0.822/0.874/0.929/0.037 ms
root@pg-primary:~#

```

IV. PostgreSQL Installation

1. Update repositories

```
sudo apt update
```

2. Install PostgreSQL

```
sudo apt install postgresql postgresql-contrib -y
psql --version
```

3. Verify Service Status

```
sudo systemctl status postgresql
```

- Enable automatic startup :
- `sudo systemctl enable postgresql`
- Verify listening ports :
- `sudo ss -tulpn | grep 5432`

```

root@pg-primary:~# systemctl status postgresql
● postgresql.service - PostgreSQL RDBMS
   Loaded: loaded (/usr/lib/systemd/system/postgresql.service; enabled; preset: enabled)
   Active: active (exited) since Fri 2026-06-12 20:16:00 UTC; 1min 53s ago
     Main PID: 15374 (code=exited, status=0/SUCCESS)
      CPU: 1ms

juin 12 20:16:00 pg-primary systemd[1]: Starting postgresql.service - PostgreSQL RDBMS...
juin 12 20:16:00 pg-primary systemd[1]: Finished postgresql.service - PostgreSQL RDBMS.
root@pg-primary:~# ss -tulpn | grep 5432
tcp        LISTEN 0      200      127.0.0.1:5432      0.0.0.0:*      users:((("postgres",pid=16268,fd=6))
root@pg-primary:~#

```

V. PostgreSQL Architecture Discovery

Switch to PostgreSQL administrator :

```
sudo -i -u postgres
```

- Access PostgreSQL shell :

```
- psql
```

- Inside PostgreSQL :

```
- SELECT version();
```

- List databases :

```
- \l
```

- List roles :

```
- \du
```

- Exit :

```
- \q
```

- Exit postgres account :

```
- exit
```

```
root@pg-primary:~# sudo -i -u postgres
postgres@pg-primary:~$ psql
psql (16.14 (Ubuntu 16.14-0ubuntu0.24.04.1))
Type "help" for help.

postgres=# SELECT version();
              version
-----
PostgreSQL 16.14 (Ubuntu 16.14-0ubuntu0.24.04.1) on x86_64-pc-linux-gnu, compiled by gcc (Ubuntu 13.3.0-6ubuntu2~24.04.1) 13.3.0, 64-bit
(1 row)

postgres=# \l
          List of databases
  Name | Owner  | Encoding | Locale Provider | Collate | Ctype  | ICU Locale | ICU Rules | Access privileges
-----|-----|-----|-----|-----|-----|-----|-----|-----
postgres | postgres | UTF8     | libc            | fr_FR.UTF-8 | fr_FR.UTF-8 |             |           | =c/postgres
template0 | postgres | UTF8     | libc            | fr_FR.UTF-8 | fr_FR.UTF-8 |             |           | postgres=CTc/postgres
template1 | postgres | UTF8     | libc            | fr_FR.UTF-8 | fr_FR.UTF-8 |             |           | =c/postgres
(3 rows)

postgres=# \du
          List of roles
 Role name | Attributes
-----|-----
 postgres | Superuser, Create role, Create DB, Replication, Bypass RLS

postgres=# \q
postgres@pg-primary:~$ exit
logout
root@pg-primary:~#
```

VI. Enterprise Database Design

We will simulate a company environment.

Database Structure

Database	Purpose
company_prod	Production
company_dev	Development
company_backup_test	Restore Testing

VII. Create Administrative Roles

1. Login

```
sudo -u postgres psql
```

2. DBA Role

```
CREATE ROLE dba_admin
```

```
WITH LOGIN
```

```
SUPERUSER
CREATEDB
CREATEROLE
PASSWORD 'StrongPassword123!';
```

3. Backup Operator

```
CREATE ROLE backup_operator
WITH LOGIN
PASSWORD 'BackupPassword123!';
```

4. Application User

```
CREATE ROLE app_user
WITH LOGIN
PASSWORD 'AppPassword123!';
```

5. Verify

```
\du
postgres=# \du
                                List of roles
Role name | Attributes
-----|-----
app_user  |
backup_operator |
dba_admin | Superuser, Create role, Create DB
postgres  | Superuser, Create role, Create DB, Replication, Bypass RLS
postgres=#
```

VIII. Create Databases

Inside PostgreSQL :

```
CREATE DATABASE company_prod;
CREATE DATABASE company_dev;
CREATE DATABASE company_backup_test;
```

Verify :

```
\l
```

```

postgres=# CREATE DATABASE company_prod;
CREATE DATABASE
postgres=# CREATE DATABASE company_dev;
CREATE DATABASE
postgres=# CREATE DATABASE company_backup_test;
CREATE DATABASE
postgres=# \l

```

Name	Owner	Encoding	Locale Provider	Collate	Ctype	ICU Locale	ICU Rules	Access privileges
company_backup_test	postgres	UTF8	libc	fr_FR.UTF-8	fr_FR.UTF-8			
company_dev	postgres	UTF8	libc	fr_FR.UTF-8	fr_FR.UTF-8			
company_prod	postgres	UTF8	libc	fr_FR.UTF-8	fr_FR.UTF-8			
postgres	postgres	UTF8	libc	fr_FR.UTF-8	fr_FR.UTF-8			
template0	postgres	UTF8	libc	fr_FR.UTF-8	fr_FR.UTF-8			=c/postgres + postgres=CTc/postgres
template1	postgres	UTF8	libc	fr_FR.UTF-8	fr_FR.UTF-8			=c/postgres + postgres=CTc/postgres

```

(6 rows)
postgres=#

```

IX. Database Ownership

Assign ownership :

```
ALTER DATABASE company_prod OWNER TO dba_admin;
```

```
ALTER DATABASE company_dev OWNER TO dba_admin;
```

```
ALTER DATABASE company_backup_test OWNER TO dba_admin;
```

Verify :

```

\l

```

```

postgres=# ALTER DATABASE company_prod OWNER TO dba_admin;
ALTER DATABASE
postgres=# ALTER DATABASE company_dev OWNER TO dba_admin;
ALTER DATABASE
postgres=# ALTER DATABASE company_backup_test OWNER TO dba_admin;
ALTER DATABASE
postgres=# \l

```

Name	Owner	Encoding	Locale Provider	Collate	Ctype	ICU Locale	ICU Rules	Access privileges
company_backup_test	dba_admin	UTF8	libc	fr_FR.UTF-8	fr_FR.UTF-8			
company_dev	dba_admin	UTF8	libc	fr_FR.UTF-8	fr_FR.UTF-8			
company_prod	dba_admin	UTF8	libc	fr_FR.UTF-8	fr_FR.UTF-8			
postgres	postgres	UTF8	libc	fr_FR.UTF-8	fr_FR.UTF-8			
template0	postgres	UTF8	libc	fr_FR.UTF-8	fr_FR.UTF-8			=c/postgres + postgres=CTc/postgres
template1	postgres	UTF8	libc	fr_FR.UTF-8	fr_FR.UTF-8			=c/postgres + postgres=CTc/postgres

```

(6 rows)
postgres=#

```

X. Create Enterprise Schema Structure

Connect :

```
\c company_prod
```

Create schemas :

```
CREATE SCHEMA hr;
```

```
CREATE SCHEMA finance;
```

```
CREATE SCHEMA inventory;
```

```
CREATE SCHEMA audit;
```

Verify :

```

SELECT schema_name
FROM information_schema.schemata ;

```



```
company_prod=# SELECT schema_name
company_prod=# FROM information_schema.schemata;
      schema_name
-----
audit
inventory
finance
hr
information_schema
pg_catalog
pg_toast
public
(8 rows)

company_prod=#
```

XI. Create Sample Tables

1. HR Table

```
CREATE TABLE hr.employees (
    id SERIAL PRIMARY KEY,
    first_name VARCHAR(100),
    last_name VARCHAR(100),
    department VARCHAR(100),
    hire_date DATE
);
```

2. Finance Table

```
CREATE TABLE finance.transactions (
    transaction_id SERIAL PRIMARY KEY,
    amount NUMERIC(10,2),
    transaction_date TIMESTAMP
);
```

3. Inventory Table

```
CREATE TABLE inventory.products (
    product_id SERIAL PRIMARY KEY,
    product_name VARCHAR(100),
    quantity INTEGER
);
```

4. Verify

```
\dt hr.*
```

```
\dt finance.*
```

```
\dt inventory.*
```

```
company_prod=# \dt hr.*
          List of relations
Schema | Name      | Type  | Owner
-----+-----+-----+-----
hr      | employees | table | postgres
(1 row)

company_prod=# \dt finance.*
          List of relations
Schema | Name      | Type  | Owner
-----+-----+-----+-----
finance | transactions | table | postgres
(1 row)

company_prod=# \dt inventory.*
          List of relations
Schema | Name      | Type  | Owner
-----+-----+-----+-----
inventory | products | table | postgres
(1 row)

company_prod=#
```

XII. Insert Test Data

1. Employees

```
INSERT INTO hr.employees
```

```
(first_name,last_name,department,hire_date)
```

```
VALUES
```

```
('John','DOE','IT','2025-01-01');
```

2. Products

```
INSERT INTO inventory.products
```

```
(product_name,quantity)
```

```
VALUES
```

```
('Laptop',20);
```

3. Transactions

```
INSERT INTO finance.transactions
```

```
(amount,transaction_date)
```

```
VALUES
```

```
(5000.00,NOW());
```

4. Verify

```
SELECT * FROM hr.employees;
```

```
SELECT * FROM inventory.products;
```

```
SELECT * FROM finance.transactions;
```

```

company_prod=# SELECT * FROM hr.employees;
 id | first_name | last_name | department | hire_date
-----+-----+-----+-----+-----
  1 | John      | DOE      | IT          | 2025-01-01
(1 row)

company_prod=# SELECT * FROM inventory.products;
 product_id | product_name | quantity
-----+-----+-----
          1 | Laptop       |        20
(1 row)

company_prod=# SELECT * FROM finance.transactions;
 transaction_id | amount | transaction_date
-----+-----+-----
              1 | 5000.00 | 2026-06-12 21:10:59.728298
(1 row)

company_prod=#

```

XIII. Permission Management

Remove public access :

```

REVOKE ALL
ON DATABASE company_prod
FROM PUBLIC;

```

Grant application access :

```

GRANT CONNECT
ON DATABASE company_prod
TO app_user;

```

Connect :

```

\c company_prod

```

Grant schema usage :

```

GRANT USAGE ON SCHEMA hr TO app_user;
GRANT USAGE ON SCHEMA inventory TO app_user;

```

Grant table access :

```

GRANT SELECT
ON ALL TABLES IN SCHEMA hr
TO app_user;

GRANT SELECT
ON ALL TABLES IN SCHEMA inventory
TO app_user;

```

Verify :

```

\dp hr.*
\dp inventory.*

```

```
company_prod=# \dp hr.employees
```

Schema	Name	Type	Access privileges	Column privileges	Policies
hr	employees	table	postgres=arwdDxt/postgres+ app_user=r/postgres		

```
(1 row)
```

```
company_prod=# \dp inventory.products
```

Schema	Name	Type	Access privileges	Column privileges	Policies
inventory	products	table	postgres=arwdDxt/postgres+ app_user=r/postgres		

```
(1 row)
```

XIV. Configure Remote Access

1. Locate Configuration

Exit PostgreSQL :

```
\q
```

Find configuration :

```
sudo -u postgres psql -c "SHOW config_file;"
```

Edit postgresql.conf :

```
sudo nano /etc/postgresql/*/main/postgresql.conf
```

Find :

```
listen_addresses = 'localhost'
```

Change :

```
listen_addresses = '*'
```

```
#-----
# CONNECTIONS AND AUTHENTICATION
#-----

# - Connection Settings -

listen_addresses = '*' # what IP address(es) to listen on;
                        # comma-separated list of addresses,
                        # defaults to 'localhost'; use '*' for all
                        # (change requires restart)
port = 5432            # (change requires restart)
max_connections = 100  # (change requires restart)
#reserved_connections = 0 # (change requires restart)
#superuser_reserved_connections = 3 # (change requires restart)
unix_socket_directories = '/var/run/postgresql' # comma-separated list of directories
                                                # (change requires restart)
#unix_socket_group = '' # (change requires restart)
#unix_socket_permissions = 0777 # begin with 0 to use octal notation
                           # (change requires restart)
#bonjour = off           # advertise server via Bonjour
                           # (change requires restart)
#bonjour_name = ''       # defaults to the computer name
                           # (change requires restart)
```

2. Configure Client Authentication

Edit :

```
sudo nano /etc/postgresql/*/main/pg_hba.conf
```

Add :

```
host all all 192.168.10.0/24 scram-sha-256
```

```
# "local" is for Unix domain socket connections only
local all all peer
# IPv4 local connections:
host all all 127.0.0.1/32 scram-sha-256
host all all 192.168.10.0/24 scram-sha-256
# IPv6 local connections:
host all all ::1/128 scram-sha-256
# Allow replication connections from localhost, by a user with the
# replication privilege.
local replication all peer
host replication all 127.0.0.1/32 scram-sha-256
host replication all ::1/128 scram-sha-256
```

Restart :

```
sudo systemctl restart postgresql
```

Verify :

```
sudo systemctl status postgresql
```

```
root@pg-primary:~# systemctl restart postgresql
root@pg-primary:~# systemctl status postgresql
● postgresql.service - PostgreSQL RDBMS
   Loaded: loaded (/usr/lib/systemd/system/postgresql.service; enabled; preset: enabled)
   Active: active (exited) since Sat 2026-06-13 12:12:28 UTC; 7s ago
     Process: 36156 ExecStart=/bin/true (code=exited, status=0/SUCCESS)
    Main PID: 36156 (code=exited, status=0/SUCCESS)
       CPU: 1ms

jun 13 12:12:28 pg-primary systemd[1]: Starting postgresql.service - PostgreSQL RDBMS...
jun 13 12:12:28 pg-primary systemd[1]: Finished postgresql.service - PostgreSQL RDBMS.
root@pg-primary:~#
```

XV. Firewall Configuration

Allow PostgreSQL traffic :

```
sudo ufw allow 5432/tcp
```

Verify :

```
sudo ufw status
```

XVI. Remote Connectivity Test

From client-test

Install client :

```
sudo apt install postgresql-client -y
```

Test :

```
psql -h pg-primary -U app_user -d company_prod
```

Password :

```
AppPassword123!
```

```

ps          ps1addtable  ps1gettable  ps1setupdate  ps1xttable  pslog      psq      pscrea      pscrea
root@client-test:~# psql -h pg-primary -U app_user -d company_prod
Password for user app_user:
psql (16.14 (Ubuntu 16.14-0ubuntu0.24.04.1))
SSL connection (protocol: TLSv1.3, cipher: TLS_AES_256_GCM_SHA384, compression: off)
Type "help" for help.

company_prod=>

```

Run :

```
SELECT * FROM hr.employees ;
```

```

company_prod=> SELECT * FROM hr.employees;
 id | first_name | last_name | department | hire_date
-----+-----+-----+-----+-----
  1 | John      | DOE      | IT          | 2025-01-01
(1 row)

company_prod=>

```

XVII. Service Administration

Useful DBA commands

- Check status
 - *sudo systemctl status postgresql*
- Restart
 - *sudo systemctl restart postgresql*
- Stop
 - *sudo systemctl stop postgresql*
- Start
 - *sudo systemctl start postgresql*
- Check logs
 - *sudo journalctl -u postgresql*